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Learning Outcomes from Data-Driven Learning

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Abstract

Since the 1960s, the use of large electronic text collections, or corpora, has evolved from educator-focused applications, such as building dictionaries and textbooks, to more learner-centred ones. This shift, exemplified by data-driven learning (DDL), empowers learners to autonomously explore linguistic inquiries using language data. This entry discusses the educational outcomes of DDL, particularly its effectiveness in second-language (L2) learning where authentic input is limited. DDL employs text-based corpora to provide authentic contexts for vocabulary and multiword units, aiding learners in recognizing language patterns and making incremental progress. However, challenges such as the relevance and comprehensibility of corpus data and the complexity of using concordancers need to be addressed, for example, by using tailored/simplified corpora and providing training opportunities for effective DDL implementation. Previous meta-analyses highlight DDL's significant benefits for writing, translation and vocabulary acquisition. Today's students are adept at using technology, such as Google, for language queries, and advances now allow for the integration of multimedia into corpora. This enhances language learning by providing textual, verbal and visual contexts. Further research is necessary to explore DDL's full potential as a language learning tool.

Keywords: Data-driven learning (DDL) · Corpus use · Second-language acquisition (SLA) · Concordancers · Learning outcome

Introduction

When large collections of electronic texts, also known as corpora (singular *corpus*), were first introduced in the 1960s, their pedagogical use was mainly focused on educators. These uses included building dictionaries and textbooks to create authentic references and instructional materials, as well as constructing learner corpora to understand language acquisition progress. Over time, however, this focus has gradually shifted to a learner-centred perspective, helping

learners address their own language learning needs (Boulton & Vyatkina, 2021). Data-driven learning (DDL) represents this shift in corpus use, enabling learners to engage in discovery learning and explore answers to their linguistic inquiries driven by language data (Johns, 1991). This chapter introduces the learning outcomes from DDL, including (1) how DDL works as a useful educational application and (2) to what extent DDL is effective in assisting language learning.

Learning Outcomes from Data-Driven Learning

DDL as a Useful Educational Application

Particularly in the context of second-language (L2) learning, where authentic language input is relatively limited in and outside classrooms compared to its first language (L1) counterpart, DDL has been widely applied as an effective learning approach. Traditionally, whether containing written or spoken language, text-based corpora are predominantly used for this purpose as they provide learners with authentic textual contexts of target vocabulary items. Therefore, corpus use includes not only single vocabulary items but also multiword units, such as collocations, presented in a Keyword in Context (KWIC) format. In this format, a large number of example sentences of the target item (i.e. concordance lines) are selected from corpora. According to Boulton and Cobb's (2017) review, consulting these language data helps learners (1) "recognize the fuzzy nature of authentic language use in context" (p. 350) and (2) detect linguistic patterns and acquire rules by making incremental progress towards the target norm.

Of course, DDL is not a panacea, and some caveats must be addressed to ensure its pedagogical benefits. First, the language data extracted from corpora may not be relevant or provide useful information about the target vocabulary items nor be comprehensible to the learners (e.g. Brezina & Gablasova, 2015). Second, learning how to use concordancers and navigating concordance lines may be laborious for learners, because these lines are not originally designed for gaining meaning like traditional linear reading materials. To remedy these limitations, according to Lee et al.'s (2019) review, researchers have suggested providing tailored or simplified corpora to ensure the language data is suitable for learners' needs (e.g. Wild et al., 2012) and offering training opportunities for corpus consultation skills prior to their initial use of concordances to enhance learners' understanding of how to navigate concordance lines and their skills in utilizing corpora for discovery learning.

Pedagogical Effectiveness of DDL from Findings of Meta-Analyses

Recent meta-analyses, such as those conducted by Boulton and Cobb (2017), Lee et al. (2019), Ueno and Takeuchi (2023) and Ngo and Chen (2024), offer valuable insights into the effectiveness of DDL in language learning. Based on 88 independent samples from 64 studies, Boulton and Cobb's meta-analysis evaluated DDL's pedagogical impact across five language skills (listening, speaking, reading, writing and translation) and four language components (vocabulary, lexicogrammar, grammar and discourse). While they found insufficient evidence to support DDL's efficacy in enhancing speaking and listening due to a lack of relevant studies, they observed significant benefits for writing and translation. Regarding language components, their findings indicate that DDL is particularly effective for vocabulary acquisition, understanding word usage in context, improving knowledge of collocations and colligations and enhancing discourse and textual awareness. Furthermore, Boulton and Cobb found that the pedagogical effectiveness of DDL was influenced by several factors. For example, DDL was more effective in foreign language (FL) learning contexts than in second-language (SL) learning contexts. Regarding interaction types, they found that DDL was more effective when used by students themselves (e.g. using concordancers or CALL (computer-assisted language learning) programmes for their linguistic inquiries) rather than when receiving learning materials designed via DDL. Lastly, its pedagogical effectiveness was greater when tailor-made local corpora were used compared to large public corpora.

As a follow-up meta-analysis to Boulton and Cobb (2017), Ueno and Takeuchi (2023) collected 422 studies to synthesize DDL's overall effects on L2 learning retention. They identified that DDL aids in the short-term retention of L2 learning but offers limited support for long-term retention. In terms of pedagogical effectiveness, Ueno and Takeuchi found that DDL is effective for interpreting, reading, writing and vocabulary and collocations but not for translation. With regard to moderators, they identified findings similar to those of Boulton and Cobb (2017), such as DDL being more effective in FL than SL contexts. They also found that using concordancers was less effective when utilized in a paper-based format.

For specific target skills, Lee et al.'s (2019) meta-analysis, based on 28 independent samples from 29 studies, focused particularly on DDL's effectiveness on vocabulary learning. Using precise knowledge, in-depth knowledge and productive use ability as dimensions of L2 vocabulary knowledge, they identified that DDL was most effective in enhancing learners' in-depth knowledge (i.e. "being able to state the referential meaning as well as syntactic features of a target lexical item", p. 726), followed by productive use ability and precise knowledge. Similar to previously introduced meta-analyses, Lee et al. also conducted moderator analyses to identify which factors influence the overall pedagogical effectiveness of DDL, particularly on L2 vocabulary learning. Their results indicated that DDL was less effective for learners

with lower L2 proficiency. Also, similar to previous findings, they identified that DDL was less effective when used in a paper-based format. Further, they found that providing training opportunities prior to DDL activities was more effective.

While Lee et al.'s study focused on vocabulary learning, Ngo and Chen's (2024) meta-analysis focused on learners' writing performance. Based on 30 studies, they found that DDL was particularly more effective for improving learners' writing at the lexicogrammatical and phraseological levels compared to the discourse and structural levels. Both Lee et al. and Ngo and Chen attributed these similar findings across vocabulary and writing components to the fact that concordance lines, which consist of words or phrases, are the fundamental unit for corpus consultation during DDL. When it comes to their moderator analysis, they found that DDL was most effective for learners with high L2 proficiency. However, they found that hands-on and hands-off DDL approaches had both large effect sizes, unlike previous findings. Further, they found that both general corpora and tailored local corpora exhibited large effect sizes, which is, again, unexpected when compared to previous findings.

Taken together, previous meta-analyses agree on DDL's positive pedagogical effectiveness as a useful education application. When it comes to more details, although the findings from moderator analyses look similar in general, such that DDL is more effective for learners with higher L2 proficiency, we could identify some discrepancies as well, such as using tailored local corpora as hands-on activity using concordancers and CALL programmes may not always be more effective. Not only do most of these findings came from subgroup analysis (Boulton & Cobb, 2017; Ngo & Chen, 2024; Ueno & Takeuchi, 2023) not from multiple meta-regression (Lee et al., 2019), but also meta-analyses cannot be always successful in encompassing all the differences across primary studies due to practical limitations. Therefore, more studies on DDL should be conducted to empirically validate such discrepancies, which will still contribute to the DDL literature.

Future Direction

Today's students are already familiar with DDL, such as in "searching for answers to their language questions, especially via the use of Google as a concordancer for the Web as corpus" (p. 351, Boulton & Cobb, 2017). Furthermore, today's technology has no limitation in incorporating multimedia materials into corpora, allowing learners to retrieve online (e.g. YouTube) and offline multimedia authentic sources for their DDL. This means that not only textual contexts but also verbal and visual contexts can be provided for more effective language learning as part of DDL. Taken together, the learning outcomes from DDL can be exponentially extended considering these changes. More research is needed to broaden our understanding of DDL and its possibilities as an effective language learning tool.

Cross-References

► AI Chatbots and Data-Driven Learning
► Future Challenges and Opportunities for Data-Driven Learning
► Learner Attitudes, Styles, Strategies and Behaviours in Data-Driven Learning
► (Lexico)grammar and Data-Driven Learning
► Pedagogical Corpora and Data-Driven Learning
► Pedagogical Guides for Data-Driven Learning
► Sustainable and Inclusive Data-Driven Learning Practices
► Teacher Education for Pedagogical Uses of Corpora
► Techniques for Beginner-Level Data-Driven Learning
► Tools for Data-Driven Learning

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